Inventory Planning and Replenishment Policy

Policy ID: SC-POL-001 | Version: 1.0 | Effective: 15 January 2026

Policy owner	Director of Supply Chain
Applies to	All stocked finished goods and critical components
Review cycle	Annual, or after a material supply disruption
Document status	Approved for internal use

1. Purpose

This policy defines how Northstar Components plans inventory, identifies stockout risk, and approves replenishment. Its objective is to protect customer service while avoiding unnecessary working capital.

2. Product classifications

Every stocked item must be assigned one of three service classes. Critical items can stop production or prevent delivery of a contracted customer order. Standard items support normal operations but have viable substitutes. Non-critical items have limited operational impact and may be replenished through routine purchasing cycles.

Service class	Safety stock requirement	Planning priority
Critical	Greater of 10 days of average demand or 100 units	Highest
Standard	7 days of average demand	Normal
Non-critical	3 days of average demand	Routine

3. Replenishment calculations

Average daily demand must be calculated from the most recent 90 days of fulfilled demand. Documented exceptional events may be excluded only with approval from the Inventory Planning Manager.

Reorder point = (average daily demand x supplier lead time in days) + safety stock.

A replenishment review must be triggered when the inventory position is at or below the reorder point. Inventory position equals on-hand stock plus confirmed inbound quantity minus allocated customer demand.

Recommended order quantity = target stock - inventory position. The result must be rounded up to the supplier minimum order quantity. Target stock covers lead-time demand, safety stock, and the next review cycle.

4. Stockout risk

An item is at risk of stockout when projected available inventory before the next confirmed delivery falls below its required safety stock. A critical item with fewer days of cover than the fastest approved supplier lead time must be classified as high risk.

High-risk critical items must be reviewed on the same business day. The planner must compare approved suppliers, open purchase orders, transport options, minimum order quantities, and unit cost before recommending action.

5. Purchase approval thresholds

The AI assistant may prepare a recommendation and a draft purchase order, but it must never create, approve, transmit, or modify a purchase order without an explicit human approval recorded by the application.

Total order value	Minimum approver
Up to EUR 10,000	Procurement Manager
EUR 10,000.01 to EUR 50,000	Director of Supply Chain
Above EUR 50,000	Chief Operating Officer

Splitting one requirement into multiple orders to avoid an approval threshold is prohibited. Emergency purchases still require the designated approver, although approval may be recorded after verbal authorization when a production shutdown is imminent.

6. Data quality and exceptions

Recommendations must identify missing or stale inputs. Inventory snapshots older than 24 hours, supplier lead times not reviewed within 180 days, and unconfirmed inbound orders must be disclosed to the decision maker.

The assistant must cite the policy section and operational records used. If required evidence is unavailable or contradictory, it must ask for human review rather than presenting a definitive recommendation.

7. Roles and responsibilities

Inventory Planner: monitors exceptions and prepares replenishment recommendations. Procurement Manager: validates supplier and commercial terms. Supply Chain Director: owns this policy and approves material exceptions. System Owner: maintains access controls, logs, and model evaluation records.

End of policy - SC-POL-001 version 1.0